

# Radiation-Hydrodynamics Simulation

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## 1 Hydrodynamics scheme

The scheme uses the principle of operator splitting, where the hydrodynamics and radiation are solved separately. The hydrodynamics is solved using a finite volume Lagrangian method (e.g. Richtmyer-Morton) and the radiation is solved using a finite difference method. The coupling between the two is done through the source terms in the hydrodynamics equations, which are updated after each radiation step.

### 1.1 Hydrodynamics step

The following uses the Euler equations:

$$\frac{\partial u}{\partial t} = -\frac{\partial P}{\partial m} \quad (1)$$

$$\frac{\partial V}{\partial t} = \frac{\partial u}{\partial m} \quad (2)$$

$$\frac{\partial e}{\partial t} = -P \frac{\partial u}{\partial m} \quad (3)$$

$$p = r\rho e \quad (4)$$

$$P = p + q \quad (5)$$

where  $E_{\text{rad}}$  is the energy exchange with the radiation field, which is updated after the radiation step (kept aside for now). In the lagrangian scheme, we begin with a certain pressure that applies a force on the mass elements, which causes them to accelerate. The scheme goes as follows:

1. Compute acceleration from the pressure gradient:

$$a_i^n = -2 \cdot \frac{\left(p_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^n\right) - \left(p_{i-\frac{1}{2}}^n + q_{i-\frac{1}{2}}^n\right)}{m_{i+\frac{1}{2}} + m_{i-\frac{1}{2}}} \quad (6)$$

2. Update velocity using the acceleration:

$$u_i^{n+\frac{1}{2}} = u_i^n + a_i^n \Delta t \quad (7)$$

3. Update node positions using the updated velocities (which is exactly a helper step for updating the volume):

$$x_i^{n+1} = x_i^n + u_i^{n+\frac{1}{2}} \Delta t \quad (8)$$

4. Update the volume and density of each cell using the new node positions:

$$V_{i+\frac{1}{2}}^{n+1} = \frac{x_{i+1}^{n+1} - x_i^{n+1}}{m_{i+\frac{1}{2}}} \quad (9)$$

$$\rho_{i+\frac{1}{2}}^{n+1} = \frac{1}{V_{i+\frac{1}{2}}^{n+1}} \quad (10)$$

5. Calculate artificial viscosity  $q$  to handle shocks, using the von Neumann-Richtmyer prescription:

$$q_{i+\frac{1}{2}}^{n+1} = \begin{cases} \sigma \rho_{i+\frac{1}{2}}^{n+1} (u_{i+1}^{n+1} - u_i^{n+1})^2 & \text{if } u_i^{n+1} > u_{i+1}^{n+1} \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

6. Calculate the new specific internal energy using the updated pressure and velocity:

$$e_{i+\frac{1}{2}}^{n+1} = \frac{e_{i+\frac{1}{2}}^n - \frac{1}{2} \left( p_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^n + q_{i+\frac{1}{2}}^{n+1} \right) \left( \frac{V_{i+\frac{1}{2}}^{n+1} - V_{i+\frac{1}{2}}^n}{m_{i+\frac{1}{2}}} \right)}{1 + \frac{\gamma-1}{2} \rho_{i+\frac{1}{2}}^{n+1} \frac{V_{i+\frac{1}{2}}^{n+1} - V_{i+\frac{1}{2}}^n}{m_{i+\frac{1}{2}}}} \quad (12)$$

7. Calculate the new pressure using the equation of state:

$$p_{i+\frac{1}{2}}^{n+1} = (\gamma - 1) \rho_{i+\frac{1}{2}}^{n+1} e_{i+\frac{1}{2}}^{n+1} \quad (13)$$

8. Compute acceleration from the **new** pressure gradient:

$$a_i^{n+1} = -2 \cdot \frac{\left( p_{i+\frac{1}{2}}^{n+1} + q_{i+\frac{1}{2}}^{n+1} \right) - \left( p_{i-\frac{1}{2}}^{n+1} + q_{i-\frac{1}{2}}^{n+1} \right)}{m_{i+\frac{1}{2}} + m_{i-\frac{1}{2}}} \quad (14)$$

9. Update velocity using the new acceleration:

$$u_i^{n+1} = u_i^{n+\frac{1}{2}} + a_i^{n+1} \frac{\Delta t}{2} \quad (15)$$

and go back to step 2 for the next iteration. **Make sure to understand the reason for the half time step in the velocity update - it makes the scheme second-order and less noisy.**

## 1.2 Coupling with radiation

This is all good for the hydrodynamics step, but we need to couple it with radiation! After step 9, we define:

$$e^* = e^{n+1} \quad (16)$$

which means that the energy isn't updated yet! we then assume that  $U_m^n = \rho e^*$  is the energy density of the matter.

# 2 Radiation Scheme

## 2.1 Physical PDEs

1. Writing the diffusion equation for the radiation energy  $E = aT_r^4$  and the effective material radiative energy  $U_R = aT_m^4$  (the energy that the material, modeled as a black body, would radiate outwards):

$$\frac{\partial E}{\partial t} + \rho \frac{\partial F}{\partial m} = \chi c \sigma (U_R - E) \quad (17)$$

$$\frac{\partial U_m}{\partial t} = -\chi c \sigma (U_R - E) \Rightarrow \frac{1}{\beta} \frac{\partial U_R}{\partial t} = -\chi c \sigma (U_R - E) \quad (18)$$

$$F = -D\rho \frac{\partial E}{\partial m} \quad \left( D = \frac{c}{3\sigma(T, \rho)} \right) \quad (19)$$

In the equation here, one assumes that during the radiation substep we keep  $\rho^*$  (to be defined later) fixed, and then linearize  $U_R(U_m)$  locally around  $U_m^*$  to get the relation  $\frac{1}{\beta} \frac{\partial U_R}{\partial t} = -\chi c \sigma (U_R - E)$ , where  $\beta = \frac{dU_R}{dU_m}|_{U_m^*}$  is the coupling coefficient between the radiation and matter that satisfies  $dU_R \approx \beta dU_m$  around  $U_m^*$ .

2. The self similar relations for the opacity (and thus the diffusion constant):

$$\frac{1}{\kappa_R} = gT^\alpha \rho^{-\lambda} \quad (20)$$

$$\frac{1}{\sigma(T, \rho)} = \frac{1}{\rho \kappa_R(T)} = gT^\alpha \rho^{-\lambda-1} \Rightarrow D = \frac{c}{3} gT^\alpha \rho^{-\lambda-1} \quad (21)$$

and the self similar relation for the internal energy:

$$e = fT^\gamma \rho^{-\mu} \quad (22)$$

where  $e$  is the specific internal energy, which means the material energy density is:

$$U_m(T) = \rho e = fT^\gamma \rho^{-\mu+1} \quad (23)$$

3. Notice that we've converted the space derivatives to mass derivatives, using the relation:

$$\frac{\partial}{\partial x} = \frac{\partial m}{\partial x} \frac{\partial}{\partial m} = \rho \frac{\partial}{\partial m} \quad (24)$$

as done in the Appendix.

4. To sum up, we have **a system of PDEs over  $T_m$  and  $T_r$** :

$$\frac{\partial(aT_r^4)}{\partial t} - \rho \frac{\partial}{\partial m} \left( D \rho \frac{\partial(aT_r^4)}{\partial m} \right) = \chi c \sigma a (T_m^4 - T_r^4) \quad (25)$$

$$\frac{1}{\beta} \frac{\partial(aT_m^4)}{\partial t} = -\chi c \sigma a (T_m^4 - T_r^4) \quad (26)$$

The system is solved over a domain of  $N$  cells with BC of ... **to be defined**.

## 2.2 Finite Difference Scheme

### 2.2.1 Discretization of a lagrangian PDE

1. In the diffusion equation, we have terms like:

$$\frac{\partial E}{\partial x} = \rho \frac{\partial E}{\partial m} \quad (27)$$

**but how are they being discretized?** We can discretize them by simply writing (leaving  $\rho$  outside of discussion for now):

$$\frac{\partial E}{\partial m} \approx \frac{E_{i+\frac{1}{2}}^n - E_{i-\frac{1}{2}}^n}{\Delta m_i} \quad (28)$$

$$\Delta m_i := \frac{1}{2}((m_{i+1} - m_i) + (m_i - m_{i-1})) = \frac{1}{2}(\Delta m_{i+\frac{1}{2}} + \Delta m_{i-\frac{1}{2}}) \quad (29)$$

where the lagrangina coordinate  $m_i$  for  $i = 0, \dots, N$  is defined as the mass of the material up to the vertex located at  $x_i$ . Therefore  $m_i = \sum_{j=0}^i \Delta m_{i+\frac{1}{2}}$  where  $\Delta m_{i+\frac{1}{2}} = m_{i+1} - m_i$  is the mass of the cell  $j = i + \frac{1}{2}$  ( $j = 1, \dots, N$ ) between the vertices  $i$  and  $i + 1$ .

2. It may also be denoted  $\Delta m_i = m_{i+\frac{1}{2}} - m_{i-\frac{1}{2}}$  where  $m_{i+\frac{1}{2}} = m_i + \frac{1}{2}\Delta m_{i+\frac{1}{2}}$  and  $m_{i-\frac{1}{2}} = m_{i-1} + \frac{1}{2}\Delta m_{i-\frac{1}{2}}$ .

$$\begin{aligned} \Delta m_i &= m_{i+\frac{1}{2}} - m_{i-\frac{1}{2}} \\ &= \left(m_i + \frac{1}{2}\Delta m_{i+\frac{1}{2}}\right) - \left(m_{i-1} + \frac{1}{2}\Delta m_{i-\frac{1}{2}}\right) = \frac{1}{2}(\Delta m_{i+\frac{1}{2}} + \Delta m_{i-\frac{1}{2}}) \end{aligned} \quad (30)$$

3. One should also regard the problem when taking the second derivative of  $E$  with respect to  $m$  - rising from the first derivative of the flux!

$$\frac{\partial F}{\partial x} = \rho \frac{\partial F}{\partial m} = \rho \frac{\partial}{\partial m} \left( -D\rho \frac{\partial E}{\partial m} \right) \quad (31)$$

**What a mess! How are we supposed to discretize this?** The solution is to discretize step by step - first the discretize the first derivative of the flux, and then the first derivative of each  $E$  term.

$$\frac{\partial F}{\partial m} \approx \frac{F_{i+1}^n - F_i^n}{\Delta m_{i+\frac{1}{2}}} \quad (32)$$

and it makes sense because the flux is defined at the faces of the cells  $i + \frac{1}{2}$  whose mass is actually  $\Delta m_{i+\frac{1}{2}} = m_{i+1} - m_i$ . Then, one should discretize each of the flux terms, as we have already done for the first derivative of  $E$ :

$$\begin{aligned} F_{i+1}^n &= -\rho_{i+1}^n D_{i+1}^n \frac{E_{i+\frac{3}{2}}^n - E_{i+\frac{1}{2}}^n}{\Delta m_{i+1}} \\ F_i^n &= -\rho_i^n D_i^n \frac{E_{i+\frac{1}{2}}^n - E_{i-\frac{1}{2}}^n}{\Delta m_i} \end{aligned} \quad (33)$$

where  $\Delta m_i$ ,  $\rho_i$  and  $D_i$  terms are all explained in the previous bullets.

4. NOTE that in our case  $\Delta m_{i+\frac{1}{2}}$  is constant for all  $i$  and therefore  $\Delta m_i = \text{const}$  and is a **predefined parameter** of the simulation.

### 2.2.2 Discretization of $D$ and $\rho$ faces

Since the derivative is a property of the vertex located between two adjacent cells, we should also pay attention to the fact that  $F$ ,  $D$  and  $\rho$  should be defined at the faces of the cells (e.g. the vertices).

Since the natural definition of  $D$  and  $\rho$  is at the cell centers (we are used to terms like  $\rho_{i+\frac{1}{2}}^n$  and  $D_{i+\frac{1}{2}}^n$ ), we need to define them at the faces of the cells using some kind of averaging.

5. The most common way to average  $D$  and  $\rho$  at the faces is to use the average mean:

$$\begin{aligned} D_{i+\frac{1}{2}} &= \frac{D_i + D_{i+1}}{2} \\ \rho_{i+\frac{1}{2}} &= \frac{\rho_i + \rho_{i+1}}{2} \end{aligned} \quad (34)$$

6. In our simulation, the constant `HARMONIC_MEAN` is set by default to `False`, which means that we use the arithmetic mean to average  $D$  and  $\rho$  at the faces.

### 2.2.3 Solution of the discretized system of equations

7. We've seen that one can **eliminate the effective radiation density**  $U_R$  from the finite-difference equation for the matter energy. The discretized equation for the material energy density is:

$$\frac{1}{\beta_{i+\frac{1}{2}}^n} \frac{[U_R]_{i+\frac{1}{2}}^{n+1} - [U_R]_{i+\frac{1}{2}}^n}{\Delta t} = \chi c \sigma_{i+\frac{1}{2}}^n (E_{i+\frac{1}{2}}^{n+1} - [U_R]_{i+\frac{1}{2}}^{n+1}) \quad (35)$$

$$(1 + \beta_{i+\frac{1}{2}}^n \Delta t \chi c \sigma_{i+\frac{1}{2}}^n) [U_R]_{i+\frac{1}{2}}^{n+1} = \beta_{i+\frac{1}{2}}^n \Delta t \chi c \sigma_{i+\frac{1}{2}}^n E_{i+\frac{1}{2}}^{n+1} + [U_R]_{i+\frac{1}{2}}^n \quad (36)$$

We want to solve for  $[U_R]_{i+\frac{1}{2}}^{n+1}$ . First, we can denote  $A_{i+\frac{1}{2}}^n = \beta_{i+\frac{1}{2}}^n \Delta t \chi c \sigma_{i+\frac{1}{2}}^n$  as the coupling coefficient between the radiation and matter, which gives us:

$$(1 + A_{i+\frac{1}{2}}^n) [U_R]_{i+\frac{1}{2}}^{n+1} = A_{i+\frac{1}{2}}^n E_{i+\frac{1}{2}}^{n+1} + [U_R]_{i+\frac{1}{2}}^n \quad (37)$$

$$\boxed{[U_R]_{i+\frac{1}{2}}^{n+1} = \frac{A_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1} + \frac{1}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n} \quad (38)$$

8. Furthermore, after substituting the expressions for  $[U_R]_{i+\frac{1}{2}}^{n+1}$  &  $F_i^n$  into the discretized equation for the radiation energy  $E$ , we get a tridiagonal implicit scheme for  $E_i^{n+1}$ :

$$\boxed{a_{i+\frac{1}{2}}^n E_{i-\frac{1}{2}}^{n+1} + b_{i+\frac{1}{2}}^n E_{i+\frac{1}{2}}^n + c_{i+\frac{1}{2}}^n E_{i+\frac{3}{2}}^{n+1} = d_{i+\frac{1}{2}}^n} \quad (39)$$

where the coefficients  $a_i^n, b_i^n, c_i^n, d_i^n$  are calculated in the appendix.

9. We notice that the coefficients  $a_i^n, b_i^n, c_i^n, d_i^n$  depend on the hydrodynamics variables (e.g. density, temperature) through the diffusion constant  $D$  and the opacity  $\sigma$ . The dependency is given through Rosen's formulae for the opacity and the internal energy. First, one should find the temperature at the previous time step:

$$T_{i+\frac{1}{2}}^n = \sqrt[4]{\frac{[U_R]_{i+\frac{1}{2}}^n}{a}} \quad (40)$$

and then substitute it into the relations for  $D(T), \beta(T)$ :

$$D_{i+\frac{1}{2}}^n = \frac{c}{3} g [T_{i+\frac{1}{2}}^n]^\alpha [\rho_{i+\frac{1}{2}}^n]^{-\lambda-1} \quad (41)$$

$$\beta_{i+\frac{1}{2}}^n = \frac{dU_R}{dU_m} = \frac{4a}{f\gamma} [T_{i+\frac{1}{2}}^n]^{4-\gamma} [\rho_{i+\frac{1}{2}}^n]^{\mu-1} \quad (42)$$

$$\frac{1}{\sigma_{i+\frac{1}{2}}^n} = \frac{1}{\rho\kappa_R(T)} = g [T_{i+\frac{1}{2}}^n]^\alpha [\rho_{i+\frac{1}{2}}^n]^{-\lambda-1} \quad (43)$$

NOTE: In the coupled scheme, we will use the temperature  $T_{i+\frac{1}{2}}^*$  computed from the hydrodynamics step **directly** to calculate  $D, \beta, \sigma$  and substitute in the tridiagonal system, no need to go through  $U_R$  in the coupled scheme.

10. To sum up, the complete algorithm for the radiation scheme is as follows:
- (a) Compute the temperature at the previous time step using the effective radiation density  $U_R$ .
  - (b) Compute the diffusion constant  $D$ , the coupling coefficient  $\beta$  and the opacity  $\sigma$  using the temperature and density at the previous time step.
  - (c) Compute the coefficients  $a_i^n, b_i^n, c_i^n, d_i^n$  for the tridiagonal system.

- (d) Solve the tridiagonal system for  $E_i^{n+1}$  using the Thomas algorithm.
- (e) Update the effective radiation density  $U_R^{n+1}$  using the expression derived above.

### 3 Coupling the two schemes

1. We first perform the hydrodynamics step, which gives us the updated density  $\rho^{n+1}$  and specific internal energy  $e^{n+1}$ . We call it  $e^*$  because we haven't updated the energy with the radiation source term yet.
2. In the radiation scheme, the self-similar profiles for the opacity and internal energy depend on the material temperature. Since we know that  $U_m = \rho e^*$ , we can compute the temperature at the previous time step using the relation:

$$e_{i+\frac{1}{2}}^* = f \left[ T_{i+\frac{1}{2}}^* \right]^\gamma \left[ \rho_{i+\frac{1}{2}}^* \right]^{-\mu} \Rightarrow T_{i+\frac{1}{2}}^* = \left( \frac{e_{i+\frac{1}{2}}^*}{f} \left[ \rho_{i+\frac{1}{2}}^* \right]^\mu \right)^{\frac{1}{\gamma}} \quad (44)$$

3. Apply the radiation scheme using the temperature and density at the previous time step to compute  $\beta$ ,  $D$  and  $\sigma$  which give us the coefficients for the tridiagonal system, which we solve to get  $E_{i+\frac{1}{2}}^{n+1}$  and  $[U_R]_{i+\frac{1}{2}}^{n+1}$ .  
NOTE: The chat proposes to perform a Picard loop, where we update the temperature using the radiation energy until convergence.
4. We then update the specific internal energy after the second source term of radiation is applied:

$$e_{i+\frac{1}{2}}^{n+1} = \frac{[U_m]_{i+\frac{1}{2}}^{n+1}}{\rho_{i+\frac{1}{2}}^{n+1}} \quad (45)$$

5. Finally, we update the hydrodynamic pressure using the equation of state:

$$p_{i+\frac{1}{2}}^{n+1} = r \rho_{i+\frac{1}{2}}^{n+1} e_{i+\frac{1}{2}}^{n+1} \quad (46)$$

and go back to the hydrodynamics step for the next iteration!



## 4 Corrections

### 4.1 Correcting the scheme with black physics

1. In the radiation diffusion equation, the finite-difference form that we reached previously is:

$$LHS = \frac{\partial E}{\partial t} + \rho \frac{\partial F}{\partial m} \quad (47)$$

$$\Downarrow \quad (48)$$

$$LHS = \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i+\frac{1}{2}}^n}{\Delta t} - \frac{\rho_{i+\frac{1}{2}}^*}{m_{i+\frac{1}{2}}} \overbrace{\left( \rho_{i+\frac{3}{2}}^n \frac{2D_{i+\frac{1}{2}}^n D_{i+\frac{3}{2}}^n}{D_{i+\frac{1}{2}}^n + D_{i+\frac{3}{2}}^n} \frac{E_{i+\frac{3}{2}}^{n+1} - E_{i+\frac{1}{2}}^{n+1}}{m_{i+\frac{3}{2}}} \right)}^{-F_{i+1}^{n+1}} \\ + \frac{\rho_{i+\frac{1}{2}}^*}{m_{i+\frac{1}{2}}} \overbrace{\left( \rho_{i-\frac{1}{2}}^n \frac{2D_{i-\frac{1}{2}}^n D_{i+\frac{1}{2}}^n}{D_{i-\frac{1}{2}}^n + D_{i+\frac{1}{2}}^n} \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i-\frac{1}{2}}^{n+1}}{m_{i+\frac{1}{2}}} \right)}^{-F_i^{n+1}} \quad (49)$$

but wait! If we assume black physics, the radiation energy should be changed to  $E^*$  and not  $E^{n+1}$ . That's why the new LHS is:

$$LHS = \frac{E_{i+\frac{1}{2}}^* - E_{i+\frac{1}{2}}^n}{\Delta t} - \frac{\rho_{i+\frac{1}{2}}^*}{m_{i+\frac{1}{2}}} \overbrace{\left( \rho_{i+\frac{3}{2}}^n \frac{2D_{i+\frac{1}{2}}^n D_{i+\frac{3}{2}}^n}{D_{i+\frac{1}{2}}^n + D_{i+\frac{3}{2}}^n} \frac{E_{i+\frac{3}{2}}^{n+1} - E_{i+\frac{1}{2}}^{n+1}}{m_{i+\frac{3}{2}}} \right)}^{-F_{i+1}^{n+1}} \\ + \frac{\rho_{i+\frac{1}{2}}^*}{m_{i+\frac{1}{2}}} \overbrace{\left( \rho_{i-\frac{1}{2}}^n \frac{2D_{i-\frac{1}{2}}^n D_{i+\frac{1}{2}}^n}{D_{i-\frac{1}{2}}^n + D_{i+\frac{1}{2}}^n} \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i-\frac{1}{2}}^{n+1}}{m_{i+\frac{1}{2}}} \right)}^{-F_i^{n+1}} \quad (50)$$

2. And the RHS remains implicit, so it's the same as before:

$$RHS = \chi c \sigma (U_R - E) \quad (51)$$

$$\Downarrow \quad (52)$$

$$\begin{aligned}
RHS &= \chi c \sigma_{i+\frac{1}{2}}^n \left( \frac{A_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1} + \frac{1}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n - E_{i+\frac{1}{2}}^{n+1} \right) \\
&= \chi c \sigma_{i+\frac{1}{2}}^n \left( \frac{1}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n - \frac{1}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1} \right) \\
&= \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n - \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1}
\end{aligned} \tag{53}$$

3. Conclusion: The new LHS and RHS define a new tridiagonal system of equations for the radiation energy density  $E^*$ , which creates a new sparse matrix that needs to be solved. [Ask Shay about this conclusion - if it's correct, it requires a new algebraic derivation of the tridiagonal constants, and implementation in python.](#)
4. CORRECTION: My conclusion is incorrect.  $E^n$  should be replaced by  $E^*$  and **NOT**  $E^{n+1}$ . It makes sense because we suppose that the radiation energy should also be changed.
5. Implementation: The only coefficient that had previously (prior to the correction) included  $E^n$  is  $d^n$ . The new formula for  $d^n$  is:

$$d_i^n = \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n + \frac{1}{\Delta t} E_{i+\frac{1}{2}}^* \tag{54}$$

or in language of arrays:

$$d_j^n = \frac{\chi c \sigma_j^n}{1 + A_j^n} [U_R]_j^n + \frac{1}{\Delta t} E_j^* \tag{55}$$

## 4.2 Black (LTE) radiation scheme

1. One can also solve the rad-hydro scheme under the LTE assumption that leads to the following simpler equation:

$$\begin{aligned}
\frac{\partial E}{\partial t} + \frac{\partial F}{\partial x} &= Q \\
C_V \frac{\partial T}{\partial t} &= -Q
\end{aligned} \tag{56}$$

where we denoted:

$$\begin{aligned}
Q &= \Sigma'_a (a T_m^4 - E) \\
F &= -D \frac{\partial E}{\partial x}
\end{aligned} \tag{57}$$

$$\Downarrow \quad (58)$$

$$\begin{aligned} \frac{\partial e}{\partial t} + \frac{\partial F}{\partial x} &= 0 \\ F &= -D \frac{\partial e}{\partial x} \end{aligned} \quad (59)$$

The LTE assumption means that  $T_m = T_R = T$  and the relation between the energy densities and the temperatures is given by:

$$\begin{aligned} e &= fT^\beta \rho^{-\mu} \quad \text{Rosen's notation} \\ E &= aT^4 \quad \text{Stefan-Boltzmann law} \end{aligned} \quad (60)$$

2. The finite-difference schem for this equation is: The starting point is the coupled energy equation in LTE conditions from above:

$$\begin{aligned} \frac{\partial e}{\partial t} &= \frac{\partial}{\partial x} \left( D \frac{\partial e}{\partial x} \right) \\ \frac{\partial e}{\partial t} &= \rho \frac{\partial}{\partial m} \left( D \rho \frac{\partial e}{\partial m} \right) \end{aligned} \quad (61)$$

In the Lagrangian framework used here, the cell centers are indexed by  $i + \frac{1}{2}$ , while the cell faces (vertices) are indexed by  $i$ . In Lagrangian coordinates, the spatial derivative is written in terms of mass coordinates, using  $1/\Delta x = \rho^*/\Delta m$ , where  $\Delta m$  is the cell mass and  $\rho^*$  is the (hydro-updated) density at the cell center.

The finite-difference (implicit) scheme for the diffusion equation in Lagrangian coordinates reads:

$$\begin{aligned} LHS &= \frac{e_{i+\frac{1}{2}}^{n+1} - e_{i+\frac{1}{2}}^n}{\Delta t} \\ RHS &= \frac{1}{\Delta m_{i+\frac{1}{2}}} \left[ D_{i+1} \rho_{i+1}^* \frac{e_{i+\frac{3}{2}}^{n+1} - e_{i+\frac{1}{2}}^{n+1}}{\Delta m_{i+1}} - D_i \rho_i^* \frac{e_{i+\frac{1}{2}}^{n+1} - e_{i-\frac{1}{2}}^{n+1}}{\Delta m_i} \right] \end{aligned} \quad (62)$$

where  $D_i$  is the diffusion coefficient at face (vertex)  $i$ ,  $\Delta m_i$  is the mass of the cell between vertices  $i - 1$  and  $i$ , and  $\rho_{i+\frac{1}{2}}^*$  is the density at the cell center after the hydro update.

The harmonic mean is still used for  $D_i$ :

$$D_i = \frac{2D_{i-\frac{1}{2}}D_{i+\frac{1}{2}}}{D_{i-\frac{1}{2}} + D_{i+\frac{1}{2}}} \quad (63)$$

(harmonic mean), and similarly for  $D_{i+1}$ .

Rearranging terms to bring all unknowns  $e_{j+\frac{1}{2}}^{n+1}$  to the left yields the tridiagonal system:

$$\boxed{a_{i+\frac{1}{2}} e_{i-\frac{1}{2}}^{n+1} + b_{i+\frac{1}{2}} e_{i+\frac{1}{2}}^{n+1} + c_{i+\frac{1}{2}} e_{i+\frac{3}{2}}^{n+1} = e_{i+\frac{1}{2}}^n} \quad (64)$$

where the boxed coefficients are:

$$\boxed{a_{i+\frac{1}{2}} = -\frac{\Delta t}{\Delta m_i} \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} D_i} \quad (65)$$

$$\boxed{b_{i+\frac{1}{2}} = 1 + \frac{\Delta t}{\Delta m_{i+\frac{1}{2}}} \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} (D_i + D_{i+1})} \quad (66)$$

$$\boxed{c_{i+\frac{1}{2}} = -\frac{\Delta t}{\Delta m_{i+1}} \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} D_{i+1}} \quad (67)$$

Here, the index  $i + \frac{1}{2}$  denotes the cell center, the left neighbor is at  $i - \frac{1}{2}$  and the right neighbor at  $i + \frac{3}{2}$ , consistent with the Lagrangian grid convention.

For practical implementation, let  $j = i + \frac{1}{2}$  and define arrays for  $a_j$ ,  $b_j$ ,  $c_j$ ,  $e_j$ , etc. The resulting tridiagonal system is:

$$a_j e_{j-1}^{n+1} + b_j e_j^{n+1} + c_j e_{j+1}^{n+1} = e_j^n \quad (68)$$

which can be efficiently solved using the Thomas algorithm or any suitable tridiagonal matrix solver.

Here,  $e_{i+\frac{1}{2}}^{n+1}$  is the energy at cell center  $i + \frac{1}{2}$  at the new timestep, and the resulting system is tridiagonal and can be solved by the Thomas algorithm.

Note: If  $e$  and  $E$  are directly connected via LTE (e.g.  $e = fT^\beta \rho^{-\mu}$  and  $E = aT^4$ ) and the same equation holds for  $E$ , a similar discrete scheme applies to  $E$ .

## 5 Appendix: Derivation of the tridiagonal form coefficients in Lagrangian coordinates

1. The discretized equation for the radiation energy in Lagrangian coordinates is:

$$\frac{E_{i+\frac{1}{2}}^{n+1} - E_{i+\frac{1}{2}}^n}{\Delta t} + \rho_{i+\frac{1}{2}}^* \frac{F_{i+1}^{n+1} - F_i^{n+1}}{\Delta m_{i+\frac{1}{2}}} = \chi c \sigma_{i+\frac{1}{2}}^n ([U_R]_{i+\frac{1}{2}}^{n+1} - E_{i+\frac{1}{2}}^{n+1}) \quad (69)$$

2. Recalling that the flux and  $U_R$  are given as expressions of  $E$ , we can substitute them into the equation above to get a tridiagonal system for  $E_{i+\frac{1}{2}}^{n+1}$ .

LHS: The flux is given by:

$$F_i^n = -\rho_i^* D_i^n \frac{E_{i+\frac{1}{2}}^n - E_{i-\frac{1}{2}}^n}{\Delta m_i} \quad (70)$$

NOTE: The radiation energy density  $E$  is taken implicitly, and the D face is taken explicitly - because of numeric considerations ([I need to ask Shay about this](#)). Substituting the fluxes into the balance equation gives:

$$\begin{aligned} LHS = & \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i+\frac{1}{2}}^n}{\Delta t} - \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} \overbrace{\left( \rho_{i+1}^* D_{i+1}^n \frac{E_{i+\frac{3}{2}}^{n+1} - E_{i+\frac{1}{2}}^{n+1}}{\Delta m_{i+1}} \right)}^{-F_{i+1}^{n+1}} \\ & + \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} \overbrace{\left( \rho_i^* D_i^n \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i-\frac{1}{2}}^{n+1}}{\Delta m_i} \right)}^{-F_i^{n+1}} \end{aligned} \quad (71)$$

Rearranging the terms gives us the tridiagonal form:

$$\begin{aligned}
LHS = & \left( -\frac{\rho_{i+\frac{1}{2}}^* \rho_i^* D_i^n}{\Delta m_{i+\frac{1}{2}} \Delta m_i} \right) E_{i-\frac{1}{2}}^{n+1} \\
& + \left[ \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} \left( \frac{\rho_i^* D_i^n}{\Delta m_i} + \frac{\rho_{i+1}^* D_{i+1}^n}{\Delta m_{i+1}} \right) + \frac{1}{\Delta t} \right] E_{i+\frac{1}{2}}^{n+1} \\
& - \left( \frac{\rho_{i+\frac{1}{2}}^* \rho_{i+1}^* D_{i+1}^n}{\Delta m_{i+\frac{1}{2}} \Delta m_{i+1}} \right) E_{i+\frac{3}{2}}^{n+1} \\
& - \frac{1}{\Delta t} E_{i+\frac{1}{2}}^n
\end{aligned} \tag{72}$$

RHS: Substituting the expression for  $[U_R]_{i+\frac{1}{2}}^{n+1}$  gives:

$$\begin{aligned}
RHS = & \chi c \sigma_{i+\frac{1}{2}}^n \left( \frac{A_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1} + \frac{1}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n - E_{i+\frac{1}{2}}^{n+1} \right) \\
= & \chi c \sigma_{i+\frac{1}{2}}^n \left( \frac{1}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n - \frac{1}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1} \right) \\
= & \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n - \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1}
\end{aligned} \tag{73}$$

3. Finally, rearranging the equation  $LHS = RHS$  gives us the tridiagonal form:

$$a_{i+\frac{1}{2}}^n E_{i-\frac{1}{2}}^{n+1} + b_{i+\frac{1}{2}}^n E_{i+\frac{1}{2}}^{n+1} + c_{i+\frac{1}{2}}^n E_{i+\frac{3}{2}}^{n+1} = d_{i+\frac{1}{2}}^n \tag{74}$$

and the coefficients are given by:

$$a_{i+\frac{1}{2}}^n = -\frac{\rho_{i+\frac{1}{2}}^* \rho_i^* D_i^n}{\Delta m_{i+\frac{1}{2}} \Delta m_i} \quad (75)$$

$$b_{i+\frac{1}{2}}^n = \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} \left( \frac{\rho_i^* D_i^n}{\Delta m_i} + \frac{\rho_{i+1}^* D_{i+1}^n}{\Delta m_{i+1}} \right) + \frac{1}{\Delta t} + \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} \quad (76)$$

$$c_{i+\frac{1}{2}}^n = -\frac{\rho_{i+\frac{1}{2}}^* \rho_{i+1}^* D_{i+1}^n}{\Delta m_{i+\frac{1}{2}} \Delta m_{i+1}} \quad (77)$$

$$d_{i+\frac{1}{2}}^n = \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n + \frac{1}{\Delta t} E_{i+\frac{1}{2}}^n \quad (78)$$

where  $A_{i+\frac{1}{2}}^n = \beta_{i+\frac{1}{2}}^n \Delta t \chi c \sigma_{i+\frac{1}{2}}^n$  is the coupling coefficient between the radiation and matter.

### Two things to understand

I need to understand how the Flek & Cummings coefficient  $f$  is related to the coupling coefficient  $A$ .

## 5.1 Deeper understanding of the tridiagonal form

1. First, let's make order in the celebration of indexes. Our staggered grid is as follows:

| Start BC              |                                                                                       | Internal              | ... | Internal                      |                                                                                               | End BC                |
|-----------------------|---------------------------------------------------------------------------------------|-----------------------|-----|-------------------------------|-----------------------------------------------------------------------------------------------|-----------------------|
| $i = 0$               | $i = \frac{1}{2}$                                                                     | $i = 1$               | ... | $i = N - 1$                   | $i = N - \frac{1}{2}$                                                                         | $i = N$               |
| $j = \frac{1}{2}$     | $j = 1$                                                                               | $j = \frac{3}{2}$     | ... | $j = N - \frac{1}{2}$         | $j = N$                                                                                       | $j = N + \frac{1}{2}$ |
| node<br>$\rho_0, D_0$ | cell center<br>$E_0, [U_R]_0$<br>$\rho_{1/2}, D_{1/2}$<br>$\beta_{1/2}, \sigma_{1/2}$ | node<br>$\rho_1, D_1$ | ... | node<br>$\rho_{N-1}, D_{N-1}$ | cell center<br>$E_N, [U_R]_N$<br>$\rho_{N-1/2}, D_{N-1/2}$<br>$\beta_{N-1/2}, \sigma_{N-1/2}$ | node<br>$\rho_N, D_N$ |

where we defined  $j = i + \frac{1}{2}$  as the index for the cell centers. Now, the indexes values are:

$$\begin{aligned} i &= 0, 1, 2, \dots, N \quad (\text{nodes}) \\ j &= 1, 2, \dots, N \quad (\text{cell centers}) \end{aligned} \quad (79)$$

2. We wish to write the tridiagonal equations in a comfortable matrix form. Under the assumption that the radiation energy array  $E_j$  is defined on cell centers  $j = 1, \dots, N$  but also at  $j = 0$  and  $j = N + 1$  (ghost cells in which the value is defined via the BC), we get the follows.

For  $j = 1, \dots, N$  where  $i = j - \frac{1}{2}$ :

$$\begin{aligned}
 a_{i+\frac{1}{2}}^n E_{i-\frac{1}{2}}^{n+1} + b_{i+\frac{1}{2}}^n E_{i+\frac{1}{2}}^{n+1} + c_{i+\frac{1}{2}}^n E_{i+\frac{3}{2}}^{n+1} &= d_{i+\frac{1}{2}}^n \\
 \Downarrow \\
 a_j^n E_{j-1}^{n+1} + b_j^n E_j^{n+1} + c_j^n E_{j+1}^{n+1} &= d_j^n
 \end{aligned} \tag{80}$$

which build the following matrix system of equations:

$$\begin{bmatrix}
 a_1 & b_1 & c_1 & 0 & \cdots & 0 & 0 \\
 0 & a_2 & b_2 & c_2 & \cdots & 0 & 0 \\
 0 & 0 & a_3 & b_3 & c_3 & \cdots & 0 \\
 \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\
 0 & \cdots & 0 & 0 & a_{N-1} & b_{N-1} & c_{N-1} & 0 \\
 0 & 0 & 0 & \cdots & 0 & a_N & b_N & c_N
 \end{bmatrix}
 \begin{bmatrix}
 E_0^{n+1} \\
 E_1^{n+1} \\
 \vdots \\
 \vdots \\
 \vdots \\
 E_N^{n+1} \\
 E_{N+1}^{n+1}
 \end{bmatrix}
 =
 \begin{bmatrix}
 d_1 \\
 d_2 \\
 d_3 \\
 \vdots \\
 d_{N-1} \\
 d_N
 \end{bmatrix} \tag{81}$$

which gives us a system of  $N$  **equations with  $N + 2$  variables**.

Before we discuss the definition of the cell-centered coefficients  $a_j^n, b_j^n, c_j^n, d_j^n$  for  $j = 1, \dots, N$ , let's think of how one would implement them on a **computer program**.

#### Implementation of the coefficient calculations

There are two important factors to consider when calculating the coefficients:

- (a) How one correctly implement the cell-valued arrays and the face-valued arrays in a computer program, where there isn't a way to create half indexes or differentiate between  $i$  and  $j$  in a simple array.
- (b) The validity of the indexes of the coefficients and other variables are affected by the boundary conditions.



### 5.1.1 The matrix equation for different BCs

(a) In **Dirichlet BC** we define:

$$E_0^n = T_s(t) \quad (82)$$

$$E_{N+1}^n = 0 \quad (83)$$

so that the system of equations is now:

$$\begin{bmatrix} a_1 & b_1 & c_1 & 0 & \cdots & 0 & 0 \\ 0 & a_2 & b_2 & c_2 & \cdots & 0 & 0 \\ 0 & 0 & a_3 & b_3 & c_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 & a_{N-1} & b_{N-1} & c_{N-1} & 0 \\ 0 & 0 & 0 & \cdots & 0 & a_N & b_N & c_N \end{bmatrix} \begin{bmatrix} T_s(t) \\ E_1^{n+1} \\ \vdots \\ \vdots \\ \vdots \\ E_N^{n+1} \\ 0 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{bmatrix} \quad (84)$$

And now we are left with  $N$  equations with  $N$  unknowns! (in fact we have just added the two missing equations, that directly solve for  $E_0$  and  $E_{N+1}$ ).

At this point, the 1st and  $N$ 'th equations become:

$$a_1 T_s(t) + b_1 E_1^{n+1} + c_1 E_2^{n+1} = d_1 \quad (85)$$

$$a_N E_{N-1}^{n+1} + b_N E_N^{n+1} + c_N \cdot 0 = d_N \quad (86)$$

which can be written as:

$$b_1 E_1^{n+1} + c_1 E_2^{n+1} = d_1 - a_1 T_s(t) \quad (87)$$

$$a_N E_{N-1}^{n+1} + b_N E_N^{n+1} = d_N \quad (88)$$

So that the elements  $a_1$  and  $c_N$  are ommitted, and their effect is inserted inside the  $d$  coefficients to provide the **reduced matrix equation**:

$$\begin{bmatrix} b_1 & c_1 & 0 & \cdots & 0 \\ a_2 & b_2 & c_2 & \cdots & 0 \\ 0 & a_3 & b_3 & c_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \cdots & 0 & 0 & a_{N-1} & b_{N-1} & c_{N-1} \\ 0 & 0 & \cdots & 0 & a_N & b_N \end{bmatrix} \begin{bmatrix} E_1^{n+1} \\ \vdots \\ \vdots \\ \vdots \\ E_N^{n+1} \end{bmatrix} = \begin{bmatrix} d_1 - a_1 T_s(t) \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{bmatrix} \quad (89)$$

(b) In **Marshak BC** we define:

$$F_0^n = F(T_{bath}) \quad (90)$$

$$E_{N+1}^n = 0 \quad (91)$$

Through a derivation that appears in detail in Appendix 6 one can reduce the equation at the surface ( $i = 0$ ) into an equation of the form:

$$\tilde{b}_1 E_1^{n+1} + \tilde{c}_1 E_2^{n+1} = \tilde{d}_1 \quad (92)$$

where  $\tilde{b}_1, \tilde{c}_1, \tilde{d}_1$  are of course defined in Appendix 6. Because this new equation completely replaces the standard diffusion equation for the first cell, the ghost cell  $E_0^{n+1}$  is algebraically eliminated from our active unknowns. Therefore, the system remains exactly  $N$  equations for the  $N$  internal cells:

$$\begin{bmatrix} \tilde{b}_1 & \tilde{c}_1 & 0 & \cdots & 0 & 0 \\ a_2 & b_2 & c_2 & \cdots & 0 & 0 \\ 0 & a_3 & b_3 & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & c_{N-2} & 0 \\ 0 & \cdots & 0 & a_{N-1} & b_{N-1} & c_{N-1} \\ 0 & 0 & \cdots & 0 & a_N & b_N \end{bmatrix} \begin{bmatrix} E_1^{n+1} \\ E_2^{n+1} \\ E_3^{n+1} \\ \vdots \\ E_{N-1}^{n+1} \\ E_N^{n+1} \end{bmatrix} = \begin{bmatrix} \tilde{d}_1 \\ d_2 \\ d_3 \\ \vdots \\ d_{N-1} \\ d_N \end{bmatrix} \quad (93)$$

We are now left with exactly  $N$  equations in  $N$  unknowns, which perfectly matches the implementation size in the numerical code.

### 5.1.2 The tridiagonal coefficient calculation

$\rho, \Delta m, D, \sigma$  are all given as arrays for the cell centers (i.e.  $\rho_j = \text{rho}[j - 1]$ ) for  $j = 1, \dots, N$ . Starting from this principle, we can calculate the face-valued arrays as follows.

For  $i = 0, \dots, N$  (notice the  $[1:]$ ,  $[:-1]$  notation that forbids invalid indices):

$$\begin{aligned} \rho_i^* &= \text{avg}(\rho_{i-\frac{1}{2}}^*, \rho_{i+\frac{1}{2}}^*) \implies \text{rho\_face}[:] = \text{avg}(\text{rho}[:-1], \text{rho}[1:]) \\ D_i^n &= \text{avg}(D_{i-\frac{1}{2}}^n, D_{i+\frac{1}{2}}^n) \implies \text{D\_face}[:] = \text{avg}(\text{D}[:-1], \text{D}[1:]) \\ \Delta m_i &= \text{avg}(\Delta m_{i-\frac{1}{2}}, \Delta m_{i+\frac{1}{2}}) \implies \text{m\_face}[:] = \text{avg}(\text{m\_cells}[:-1], \text{m\_cells}[1:]) \end{aligned} \quad (94)$$

of `size=N-1`. We will define:

$$f_i^* = \frac{\rho_i^* D_i^n}{\Delta m_i} \quad (95)$$

and now we are ready to define the `flux_coeff` so that `flux_coeff.size()` is  $N - 1$ :

$$\text{flux\_coeff}[i - 1] = f_i^* \quad (96)$$

#### Face values at the first node ( $i = 0$ )

Since the size of the `flux_coeff` array is  $N - 1$ , and it should be multiplied element-wise by other  $N$  sized arrays (e.g., `lagrangian_coeff`) that are computed at the cell centers (specifically - the cell center of which the corresponding node is located at its left face since  $i = j - 1/2$ ), we shall define the value of  $\rho^*$ ,  $D^n$ ,  $\Delta m$  at  $i = 0$  by reflecting their values across the boundary, i.e.,  $f_0^* = f_1^*$ .

3. We will also define the following numeric arrays for notation comfort.  
For  $j = 1, \dots, N$ : Under the definitions:

$$\begin{aligned} \mathcal{L}_j^* &= \frac{\rho_j^*}{\Delta m_j} \\ B_j^n &= \frac{\chi c \sigma_j^n}{1 + A_j^n} \end{aligned} \quad (97)$$

The numeric arrays are:

$$\begin{aligned} \text{lagrangian\_coeff}[j-1] &= \mathcal{L}_j^* \\ \text{total\_coupling}[j-1] &= B_j^n \end{aligned} \quad (98)$$

4. To sum up, the coefficients are defined as follows.  
For  $j = 1, \dots, N$  and  $i = j - 1 = 0, \dots, N - 1$  (left face):

$$\begin{aligned} a_j^n &= -\mathcal{L}_j^* f_i^* \\ b_j^n &= \mathcal{L}_j^* (f_i^* + f_{i+1}^*) + \frac{1}{\Delta t} + B_j^n \\ c_j^n &= -\mathcal{L}_j^* f_{i+1}^* \\ d_j^n &= B_j^n [U_R]_j^n + \frac{1}{\Delta t} E_j^n \end{aligned} \quad (99)$$

and in code notation (after defining `a`, `b`, `c`, `d` arrays of `size = N`):

$$\begin{aligned}
a[j - 1] &= \mathcal{L}_j^* f_i^* \\
b[j - 1] &= \mathcal{L}_j^*(f_i^* + f_{i+1}^*) + \frac{1}{\Delta t} + B_j^n \\
c[j - 1] &= \mathcal{L}_j^* f_{i+1}^* \\
d[j - 1] &= B_j^n [U_R]_j^n + \frac{1}{\Delta t} E_j^n
\end{aligned} \tag{100}$$

#### Between code arrays and physical quantities

As stated at the beginning of this subsection, the physical quantities  $\rho, \Delta m, D, \sigma$  are all given as arrays for the cell centers (i.e.  $\rho_j = \text{rho}[j - 1]$ ) for  $j = 1, \dots, N$  so assumptions such as `arr[j-1] = f( $\rho, \Delta m, D, \sigma$ )` just mean direct array assignment!

## 6 Appendix: Boundary Conditions in the Diffusion Approximation

In the  $P_1$  diffusion approximation, the specific intensity  $I(x, \mu)$  is not tracked for every angle. Instead, we use **Marshak Boundary Conditions** to bridge the gap between averaged quantities (energy density  $E$  and flux  $F$ ) and the physical constraints at the boundaries.

### 6.1 General Physical Derivation

At a boundary interface, the net inward partial flux  $J_{in}$  must match the source drive. For a blackbody source at temperature  $T_{bath}$ ,  $J_{in} = \sigma_{sb} T_{bath}^4$ . In the diffusion approximation, the partial fluxes are related to  $E$  and  $F$  by:

$$J_{in} = \frac{cE}{4} + \frac{F}{2}, \quad J_{out} = \frac{cE}{4} - \frac{F}{2} \quad (101)$$

At the origin ( $x = 0$ ), the net flux into the material is  $F_0$ . By substituting  $\sigma_{sb} T_{bath}^4 = \frac{c}{4} E_{bath}$ , the Marshak condition becomes:

$$\frac{cE(0, t)}{4} + \frac{F_0(t)}{2} = \frac{c}{4} E_{bath}(t) \quad (102)$$

$$\boxed{F_0(t) = \frac{c}{2} E_{bath}(t) - \frac{c}{2} E(0, t)} \quad (103)$$

At the far boundary ( $x = L$ ), we assume a vacuum where no radiation enters from the right ( $J_{in, right} = 0$ ), leading to the free-streaming limit:

$$\frac{cE(L, t)}{4} - \frac{F_L(t)}{2} = 0 \implies \boxed{F_L = \frac{cE(L, t)}{2}} \quad (104)$$

### 6.2 Discretization in Lagrangian Coordinates

In our staggered grid, cell centers are indexed by  $j = 1, \dots, N$  and faces by  $i = 0, \dots, N$ . The diffusion equation at the cell located between the faces  $i$

and  $i + 1$  is:

$$\begin{aligned} \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i+\frac{1}{2}}^n}{\Delta t} - \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} \left( \overbrace{\rho_{i+1}^n D_{i+1}^n \frac{E_{i+\frac{3}{2}}^{n+1} - E_{i+\frac{1}{2}}^{n+1}}{\Delta m_{i+1}}}^{-\hat{F}_{i+1}^{n+1}} \right) + \frac{\rho_{i+\frac{1}{2}}^*}{\Delta m_{i+\frac{1}{2}}} \left( \overbrace{\rho_i^n D_i^n \frac{E_{i+\frac{1}{2}}^{n+1} - E_{i-\frac{1}{2}}^{n+1}}{\Delta m_i}}^{-\hat{F}_i^{n+1}} \right) = \\ \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} [U_R]_{i+\frac{1}{2}}^n - \frac{\chi c \sigma_{i+\frac{1}{2}}^n}{1 + A_{i+\frac{1}{2}}^n} E_{i+\frac{1}{2}}^{n+1} \end{aligned} \quad (105)$$

#### Quick note about notation

I used the notation " $\hat{F}_1$ " to indicate that the subscript denotes  $i$  and not  $j$ . For comfort indexing, in the notation " $E_1$ " (without hat) the subscript denotes  $j$  (default notation prefers cell centers).

#### First Cell ( $j = 1$ )

For the first cell, we substitute the boundary flux  $\hat{F}_0^{n+1}$  derived from the Marshak condition:

$$\frac{E_1^{n+1} - E_1^n}{\Delta t} + \frac{\rho_1^*}{\Delta m_1} \left( \hat{F}_1^{n+1} - \overbrace{\left[ \frac{c}{2} E_{bath} - \frac{c}{2} E(0, t) \right]}^{\hat{F}_0^{n+1}} \right) = S_1^{n+1} \quad (106)$$

where the source term is defined as:

$$S_1^{n+1} = \frac{\chi c \sigma_1^n}{1 + A_1^n} [U_R]_1^n - \frac{\chi c \sigma_1^n}{1 + A_1^n} E_1^{n+1} \quad (107)$$

To close the system using only internal cell unknowns, we make the standard first-order spatial approximation that the radiation energy density at the boundary face is roughly equal to the energy of the first cell:  $E(0, t) \approx E_1^{n+1}$ . Substituting this and the internal flux  $\hat{F}_1^{n+1}$  yields:

$$\begin{aligned} \frac{E_1^{n+1} - E_1^n}{\Delta t} + \frac{\rho_1^*}{\Delta m_1} \left[ \left( -\hat{\rho}_1^n \hat{D}_1^n \frac{E_2^{n+1} - E_1^{n+1}}{\Delta \hat{m}_1} \right) \right. \\ \left. - \frac{c}{2} E_{bath} + \frac{c}{2} E_1^{n+1} \right] = \\ \frac{\chi c \sigma_1^n}{1 + A_1^n} [U_R]_1^n - \frac{\chi c \sigma_1^n}{1 + A_1^n} E_1^{n+1} \end{aligned} \quad (108)$$

Notice that the surface emission term has become  $\frac{\rho_1^* c}{2\Delta m_1} E_1^{n+1}$ . This naturally moves to the LHS of the equation to maintain the implicit scheme stability. Let us define this Marshak boundary cooling coefficient  $C_1$ :

$$C_1 = \frac{c\rho_1^*}{2\Delta m_1} \quad (109)$$

Grouping terms for  $E_1^{n+1}$  and  $E_2^{n+1}$  yields the modified tridiagonal system:

$$\begin{aligned} & \left( \frac{1}{\Delta t} + \frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1} + C_1 + \frac{\chi c \sigma_1^n}{1 + A_1^n} \right) E_1^{n+1} \\ & - \left( \frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1} \right) E_2^{n+1} = \\ & \frac{E_1^n}{\Delta t} + C_1 E_{bath} + \frac{\chi c \sigma_1^n}{1 + A_1^n} [U_R]_1^n \end{aligned} \quad (110)$$

where  $E_{bath} = aT_{bath}^4(t)$ .

This gives us the modified tridiagonal coefficients for the first cell:

$$\boxed{\tilde{b}_1^n = \frac{1}{\Delta t} + \frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1} + C_1 + \frac{\chi c \sigma_1^n}{1 + A_1^n}} \quad (111)$$

$$\boxed{\tilde{c}_1^n = -\frac{\rho_1^* \hat{\rho}_1^n \hat{D}_1^n}{\Delta m_1 \Delta \hat{m}_1}} \quad (112)$$

$$\boxed{\tilde{d}_1^n = \frac{E_1^n}{\Delta t} + C_1 E_{bath} + \frac{\chi c \sigma_1^n}{1 + A_1^n} [U_R]_1^n} \quad (113)$$

### Relation to the base internal coefficients

We can connect these modified coefficients to the base internal coefficients  $a_1^n, b_1^n, d_1^n$  derived earlier. Recall that  $a_1^n$  represents the left-face diffusion contribution, which is  $-\frac{\rho_1^* \hat{\rho}_0^n \hat{D}_0^n}{\Delta m_1 \Delta m_0}$ , and  $b_1^n$  contains  $-a_1^n$ . The Marshak condition essentially replaces this left-face diffusion with the Marshak cooling/drive terms.

Therefore, the modified coefficients can be elegantly constructed directly from the base coefficients:

$$\boxed{\tilde{b}_1^n = b_1^n + a_1^n + C_1} \quad (114)$$

$$\boxed{\tilde{a}_1^n = 0} \quad (115)$$

$$\boxed{\tilde{d}_1^n = d_1^n + C_1 E_{bath}} \quad (116)$$

This avoids recalculating the shared terms (like the right-face flux or the source terms) and is exactly how it is implemented numerically.

**Last Cell** ( $i = N - \frac{1}{2}, j = N$ ): This is not done in our simulation because the Marshak waves don't reach the edge boundary in our simulation setup.

$$\frac{E_N^{n+1} - E_N^n}{\Delta t} + \frac{\rho_N^*}{\Delta m_N} \left( \frac{c E_N^{n+1}}{2} - F_{N-1}^{n+1} \right) = S_N^{n+1} \quad (117)$$

This modifies the final diagonal element:

$$\boxed{\tilde{b}_N^n = b_N^n + \frac{c \rho_N^*}{2 \Delta m_N}} \quad (118)$$

### Missing implementation details

I've not included here the implementation of the calculation of  $T_{bath}(t)$  from using Menahem's subsonic solver. What it actually does is use the semi-analytic solver to draw the



## 7 Calculation of $T_{bath}(t)$ from Menahem's subsonic solver

When solving Marshak BC problems, one gives  $T_{bath}(t)$  rather than  $T_{surface}(t)$  as the BC. In our **building of test cases**, we would like to calculate  $T_{bath}(t)$  that when used in the simulation via Marshak BC is **consistent with some other Dirichlet BC** solution.

### Derivation of driving radiation bath temperature

At the boundary ( $m = 0$ ), the net radiation energy flux entering the system is governed by the physical Marshak boundary condition:

$$F(0, t) = \sigma_{sb} [T_{bath}^4(t) - T_s^4(t)] \quad (119)$$

Given the power-law surface material temperature boundary condition,  $T_s(t) = T_0 t^\tau$ , this yields:

$$T_{bath}(t) = \left[ T_0^4 t^{4\tau} + \frac{F(0, t)}{\sigma_{sb}} \right]^{1/4} \quad (120)$$

From the subsonic similarity formulation (Eq. 638), the physical flux boundary value is matched to the self-similar dimensional radiation energy flux at the origin ( $\xi = 0$ ),  $F(0, t) = s(0, t)$ :

$$F(0, t) = A^{a+(q+1)a_1+na_3} B^{1+b+(q+1)b_1+nb_3} t^{c+(q+1)c_1+nc_3} S(0) \quad (121)$$

By fully substituting the structural expressions for the powers computed in the code:

$$\begin{aligned} n &= \frac{4 + \alpha}{\beta} \\ k &= 1 - \mu \\ q &= (1 - \mu) \left( \frac{4 + \alpha}{\beta} \right) + \lambda - 1 \\ \text{denom} &= \frac{1}{4 + 2\lambda - 4\mu} \end{aligned}$$

We write the explicit algebraic components for the exponents  $a, b, c$ ,

$a_1, b_1, c_1$ , and  $a_3, b_3, c_3$ :

$$a = \frac{2\beta - \beta\lambda - 8 - 2\alpha + \mu(4 + \alpha)}{4 + 2\lambda - 4\mu} \quad (122)$$

$$a_1 = \frac{-2(4 + \alpha - 2\beta)}{4 + 2\lambda - 4\mu} \quad (123)$$

$$a_3 = \frac{-2[\mu(4 + \alpha) - \beta\lambda - 4 - \alpha]}{4 + 2\lambda - 4\mu} \quad (124)$$

$$b = \frac{\mu - 2}{4 + 2\lambda - 4\mu} \quad (125)$$

$$b_1 = \frac{-2}{4 + 2\lambda - 4\mu} \quad (126)$$

$$b_3 = \frac{-2(\mu - 1)}{4 + 2\lambda - 4\mu} \quad (127)$$

$$c = \frac{3\mu - 2\lambda - 2 + \tau[2(\beta - \alpha - 4) - \beta\lambda + \mu(4 + \alpha)]}{4 + 2\lambda - 4\mu} \quad (128)$$

$$c_1 = \frac{-2[\tau(4 + \alpha - 2\beta) - 1]}{4 + 2\lambda - 4\mu} \quad (129)$$

$$c_3 = \frac{-2[\tau(\mu(4 + \alpha) - \beta\lambda - 4 - \alpha) + 1 - \mu]}{4 + 2\lambda - 4\mu} \quad (130)$$

Now, substituting the definitions of the dimensional constants  $A$  and  $B$ :

$$A = T_b((\gamma - 1)f)^{1/\beta}, \quad B = \frac{16\sigma_{\text{sb}}g}{3\beta((\gamma - 1)f)^n} \quad (131)$$

The cumulative explicit time-independent pre-factor is:

$$\left[ T_b((\gamma - 1)f)^{1/\beta} \right]^{a+(q+1)a_1+na_3} \left[ \frac{16\sigma_{\text{sb}}g}{3\beta((\gamma - 1)f)^n} \right]^{1+b+(q+1)b_1+nb_3} S(0) \quad (132)$$

Substituting this complete expansion back into the Marshak expression, we isolate  $T_0 t^\tau$  outside the bracket to reveal the explicit non-equilibrium boundary correction term:

$$T_{\text{bath}}(t) = T_0 t^\tau [1 + \mathcal{M}_{\text{explicit}} \cdot t^{\Delta\eta}]^{1/4} \quad (133)$$

Where the fully expanded constant scalar  $\mathcal{M}_{\text{explicit}}$  is defined by:

$$\mathcal{M}_{\text{explicit}} = \frac{S(0)}{\sigma_{\text{sb}} T_0^4} \left[ T_b((\gamma - 1)f)^{1/\beta} \right]^{a+(q+1)a_1+na_3} \left[ \frac{16\sigma_{\text{sb}}g}{3\beta((\gamma - 1)f)^n} \right]^{1+b+(q+1)b_1+nb_3} \quad (134)$$

And the explicit net power law deviation parameter  $\Delta\eta$  is:

$$\Delta\eta = c + (q+1)c_1 + nc_3 - 4\tau \quad (135)$$

$$= \frac{1}{4+2\lambda-4\mu} \left\{ 2 \left[ (4+\alpha)(1-\beta^{-1}) - 1 - \lambda \right] \right. \\ \left. + \tau \left[ 2\beta - \lambda(4+\alpha) - 4\alpha - 18 \right] \right\} - 4\tau \quad (136)$$

Under strict thermodynamic self-similarity where the internal drive completely tracks the external source,  $\Delta\eta = 0$ , meaning  $c + (q+1)c_1 + nc_3 = 4\tau$ . In this configuration, the temporal term cancels from the bracket, delivering a constant scaling parameter over the entire timeline:

$$T_{\text{bath}}(t) = T_0 \left[ 1 + \frac{S(0)}{\sigma_{\text{sb}} T_0^4} \left[ T_b((\gamma-1)f)^{1/\beta} \right]^{a+(q+1)a_1+na_3} \left[ \frac{16\sigma_{\text{sb}}g}{3\beta((\gamma-1)f)^n} \right]^{1+b+(q+1)b_1+nb_3} \right]^{1/4} t^\tau \quad (137)$$